Projective Dynamics and Mass Hierarchy: Hierarchical Amplification via Growing Relational Valence

Jérôme Beau

Independent Researcher, Cosmochrony Project

contact@cosmochrony.org

March 2026

Abstract

The companion papers in the Admissibility Sub-Programme established two results: the representation-theoretic structure of ADE binary Cayley graphs fixes the number and ordering of stratigraphic levels [4], and the projective resolution satisfies $\Lambda_{\mathrm{proj}}(n) \asymp \lambda_2(n)$ under the isoperimetric closure hypothesis [5]. In the Lubotzky–Phillips–Sarnak (LPS) graph model at fixed prime p, the algebraic connectivity $\lambda_2(n)$ converges to a constant, making Λ_{proj} asymptotically static. This static regime produces mass ratios in [0.10, 2.2], far below the 10^{-3}–10^{-5} range observed in the Standard Model, and inverts the mass ordering expected from the admissibility envelope.

We show that both failures are resolved by allowing the effective relational valence p to grow with the cascade rank n. As $p(n) \to \infty$, the Kesten–McKay spectral support $[\lambda_-(p), \lambda_+(p)]$ narrows toward the midpoint $\lambda = 1$. The three fixed ADE eigenvalue levels exit the support in a rank-dependent order determined by their distance from 1: the level farthest from 1 exits first and therefore stabilises at the highest mass. For the physically relevant ADE substrates this restores the admissibility ordering $M_3 > M_2 > M_1$ without additional tuning. The exit ranks satisfy an implicit equation $\lambda_i = \lambda_+(p(n_{\mathrm{exit}}(\lambda_i)))$, which for a power-law growth $p(n) \sim n^\beta$ yields mass ratios that grow rapidly with β, providing a single falsifiable structural parameter.

Keywords: spectral graph theory; Ramanujan graphs; Kesten–McKay distribution; projective resolution; mass hierarchy; relational valence.

Contents

1 Introduction

1.1 Context and logical chain

The Cosmochrony Bounded Admissibility framework [1] associates to each cascade rank n an energy scale $E_n = E_{\mathrm{P}}/n$ and a projective resolution $\Lambda_{\mathrm{proj}}(n)$: a spectral mode of eigenvalue λ is admissible at rank n if and only if $\lambda \leq \Lambda_{\mathrm{proj}}(n)$. The stabilisation rank of a mode is the first n at which it ceases to be admissible, and the corresponding energy is the proxy for the physical mass of the associated particle sector.

Four preceding papers in the Admissibility Sub-Programme have progressively constrained this picture. Step 1 [2] established that the bounded-flux constraint selects spin-$\frac{1}{2}$ sectors as spectrally dominant on Cayley graphs of SU(2) subgroups. Step 2 [3] identified the binary icosahedral group $2I$ as the unique maximiser of the spectral capacity functional. Step 3 [4] showed that exactly four pairs (G, S) from the ADE binary groups produce Cayley graphs with exactly three distinct non-zero eigenvalues, yielding a three-level stratigraphic structure compatible with three fermion generations, and proved a no-go theorem ruling out scalar calibration as the source of level discreteness. Step 4 [5] derived the isoperimetric law $\Lambda_{\mathrm{proj}}(n) \asymp h(G(n))^2$ and, under a spectral-isoperimetric saturation hypothesis, the linear law $\Lambda_{\mathrm{proj}}(n) \asymp \lambda_2(n)$. It then evaluated the resulting mass ratios in the LPS model at fixed prime p and identified both the ordering inversion and the amplitude failure as consequences of the asymptotic constancy of Λ_{proj} in that regime. Step 5a [6] establishes, in a companion paper written concurrently with the present work, that allowing p to grow with the cascade rank removes the fixed-valence saturation and restores the correct mass ordering $M_3 > M_1 > M_2$ through the support-narrowing mechanism. That paper explicitly leaves open the quantitative determination of the inter-generation mass ratios, which is the central result of the present paper.

The present paper is Step 5 of the sub-programme. Its starting point is the open direction (O1) identified in [5]: allow the effective relational valence p to grow with the cascade rank.

1.2 What the static LPS regime established and where it fails

In the LPS model with fixed prime p, the algebraic connectivity satisfies

$$\lambda_2(n) \;\longrightarrow\; \lambda_2^*(p) \;:=\; 1 - \frac{2\sqrt{p}}{p+1} \qquad (n \to \infty), \tag{1}$$

so Λ_{proj} converges to a constant $\Lambda_0(p)$ (Proposition 4.1 of [5]). In this static-threshold regime the only active ordering mechanism is the Kesten–McKay saturation condition $N(\lambda_i; n) \geq k \dim \rho_{\lambda_i}$. The resulting mass formula is

$$\frac{M_i}{M_j} \;=\; \frac{F_{\mathrm{KM}}(\lambda_i)}{F_{\mathrm{KM}}(\lambda_j)} \cdot \frac{\dim \rho_{\lambda_j}}{\dim \rho_{\lambda_i}}, \tag{2}$$

where F_{KM} is the Kesten–McKay cumulative distribution function supported on $[\lambda_-(p), \lambda_+(p)]$ with $\lambda_\pm(p) = 1 \mp 2\sqrt{p}/(p+1)$.

As shown numerically in [5], equation (2) produces ratios in [0.10, 2.2] for all admissible primes and all three-level ADE substrates, whereas the observed inter-generation mass ratios span 10^{-3}–10^{-5} for charged leptons and up-type quarks. Furthermore, the ratio $c_3/\dim \rho_3$ exceeds $c_1/\dim \rho_1$ for the 2I (ord-4) substrate, inverting the ordering $M_3 > M_2 > M_1$ expected from the Born–Infeld admissibility envelope.

Both failures have a common structural cause: the threshold Λ_{proj} does not vary significantly along the cascade, so modes with spectrally similar eigenvalues are distinguished only by modest differences in F_{KM}, which are of order unity. Generating the observed amplification requires a mechanism through which small spectral differences translate into large differences in stabilisation rank.

1.3 The dynamic degree of freedom: $p(n) \to \infty$

The LPS construction at fixed prime p is only one point in a larger family. Physically, the prime p controls the effective relational valence of the substrate: the relaxation graph $G(n)$ is $(p+1)$-regular, so each relational node maintains $p+1$ active connections. A growing cascade need not maintain a fixed valence: as the number of active relational nodes increases, the connectivity structure of the substrate may become richer, corresponding to an effective increase of p.

Allowing p to depend on the cascade rank n introduces a new dynamical degree of freedom that was held fixed in the analysis of [5]. Its key spectral consequence is a progressive narrowing of the Kesten–McKay support:

$$\lambda_+(p) - \lambda_-(p) \;=\; \frac{4\sqrt{p}}{p+1} \;\longrightarrow\; 0 \qquad (p \to \infty). \tag{3}$$

As the support shrinks toward $\{\lambda = 1\}$, the three fixed ADE eigenvalue levels $\lambda_1 < \lambda_2 < \lambda_3$ eventually exit the support. The exit occurs in an order determined entirely by the distance of each level from the midpoint 1: the level with the largest distance exits first. Since $\lambda_3 > 1$ and for the relevant ADE substrates $\lambda_3 - 1 > 1 - \lambda_1$, the level λ_3 exits before λ_1, yielding stabilisation ranks $n_{\mathrm{exit}}(\lambda_3) < n_{\mathrm{exit}}(\lambda_1)$ and therefore masses $M_3 > M_1$, in agreement with the admissibility ordering.

This mechanism, which we call *support-narrowing amplification*, is the central object of the present paper. It does not introduce free parameters beyond the growth law $p(n)$, and the resulting mass ratios are sharply sensitive to the exponent governing that growth, providing a falsifiable structural prediction.

1.4 Structure of the paper

Section 2 collects the relevant results of [5] concerning the static LPS regime and states both failures precisely, serving as the logical starting point for the dynamic extension. Section 3 introduces the dynamic valence hypothesis $p = p(n)$ and derives the support-narrowing mechanism. Section **??** formalises the exit-rank equation and derives the resulting mass formula. Section 4 evaluates mass ratios for power-law growth $p(n) \sim n^{\beta}$ and compares with Standard Model data. Section 5 discusses the compatibility of the dynamic mechanism with the three-level ADE structure established in [4]. Section 6 provides a numerical illustration. Section **??** places O3 in the global O-series architecture. Section 7 discusses the falsifiable predictions. Section 8 concludes.

2 The Static LPS No-Go: Precise Statement

This section assembles the results of [5] that will serve as the logical departure point for the dynamic extension. No new results are proved here; the purpose is to state the two failures of the static regime with the precision required to identify their structural cause.

2.1 Setup and notation

Throughout this paper, $n \in \mathbb{N}_{>0}$ denotes the *cascade rank*, an integer-valued depth parameter indexing successive stages of the Cosmochrony relaxation cascade. The energy available at rank n is $E_n = E_{\mathrm{P}}/n$, where E_{P} is the Planck energy. We write $G(n)$ for the effective relaxation graph at rank n: a connected, $d(n)$-regular, undirected graph. Its normalised Laplacian is $L_{G(n)}$, with eigenvalues $0 = \mu_0 < \mu_1 \le \mu_2 \le \cdots$. The *algebraic connectivity* (Fiedler value) is $\lambda_2(n) := \mu_1(L_{G(n)})$.

The projective resolution is

$$\Lambda_{\mathrm{proj}}(n) \;=\; \frac{c_\chi(n)^2}{A_{\min}^2}, \tag{4}$$

where $c_\chi(n) > 0$ is the global projective flux bound and $A_{\min} > 0$ is a fixed scale independent of n. Under the isoperimetric closure hypothesis (Hypothesis 2.1 of [5]), $c_\chi(n) \asymp h(G(n))$, giving $\Lambda_{\text{proj}}(n) \asymp h(G(n))^2$. Under the additional spectral-isoperimetric saturation hypothesis (Hypothesis 2.2 of [5]), the linear law

$$\Lambda_{\text{proj}}(n) \;\asymp\; \lambda_2(n) \tag{5}$$

holds with constants independent of n.

The LPS graph $X^{p,q}$ is a $(p+1)$-regular Ramanujan graph on $\mathrm{PSL}(2, \mathbb{F}_q)$, with $|V(X^{p,q})| = \frac{1}{2}q(q^2-1)$. The physical identification $n \sim |V(G(n))|$ gives $n \sim q^3/2$, so $q \sim (2n)^{1/3}$. The Ramanujan property gives the uniform lower bound

$$\lambda_2(X^{p,q}) \;\geq\; 1 - \frac{2\sqrt{p}}{p+1} \qquad \text{for all admissible } q. \tag{6}$$

2.2 Asymptotic constancy of the projective threshold

The following result is Proposition 4.1 of [5].

Proposition 2.1 (Asymptotic constancy of Λ_{proj}, [?])**.** *Let p be a fixed prime. Under the linear law* (5)*, the projective resolution satisfies*

$$\Lambda_{\text{proj}}(n) \;\longrightarrow\; \Lambda_0(p) \;:=\; \frac{C\,\lambda_2^*(p)}{A_{\min}^2}, \qquad \lambda_2^*(p) = 1 - \frac{2\sqrt{p}}{p+1}, \qquad (n \to \infty), \tag{7}$$

where $C > 0$ absorbs the implicit constant in (5)*. In particular, Λ_{proj} is asymptotically static: its variation along the cascade is suppressed as $n^{-1/3}$.*

When Λ_{proj} is asymptotically constant, the support condition $\lambda_i \leq \Lambda_{\text{proj}}(n)$ reduces to a fixed constraint for all sufficiently large n, and the sole ordering mechanism becomes the saturation condition $N(\lambda_i; n) \geq k \dim \rho_{\lambda_i}$.

2.3 The Kesten–McKay mass formula and its two failures

For the three-level ADE substrates identified in [4], the Kesten–McKay cumulative distribution function $F_{\text{KM}}(\lambda) = \mu_{\text{KM}}^{(p)}((-\infty, \lambda])$ controls the linear growth of the spectral counting function: $N(\lambda; n) \sim F_{\text{KM}}(\lambda) \cdot n$ as $n \to \infty$ (Proposition 4.2 of [5]). The stabilisation rank is therefore $n_{\text{proj}}(\lambda_i) \sim k \dim \rho_{\lambda_i} / F_{\text{KM}}(\lambda_i)$, giving the mass formula (2).

Two structural failures of this formula are established in Section 5 of [5].

Proposition 2.2 (Amplitude failure)**.** *For all three three-level ADE substrates and all admissible primes, the predicted mass ratios satisfy*

$$0.10 \;\leq\; \frac{M_i}{M_j} \;\leq\; 2.2. \tag{8}$$

The observed inter-generation ratios for charged leptons ($m_e/m_\tau \approx 2.9 \times 10^{-4}$) and up-type quarks ($m_u/m_t \sim 10^{-5}$) fall short of the predicted range by two to four orders of magnitude.

Proposition 2.3 (Ordering failure for $2I$ (ord-4))**.** *For the $2I$ substrate with order-4 generators, the ratio $c_3(p)/\dim \rho_3$ exceeds $c_1(p)/\dim \rho_1$ for all admissible primes $p \leq 29$, predicting $M_1 < M_3$ in contradiction with the admissibility ordering $M_1 > M_3$ (lower eigenvalue $\Leftrightarrow$ wider admissibility window $\Leftrightarrow$ heavier particle) established in [4].*

Remark 2.4 (Common structural cause). *Both failures share the same structural origin. In the static-threshold regime, all three ADE levels simultaneously satisfy the support condition* $\lambda_i \leq \Lambda_0(p)$ *for all sufficiently large* n*. Level discrimination therefore relies entirely on differences in* $F_{\mathrm{KM}}(\lambda_i)$*, which are of order unity by the boundedness of* F_{KM}*. No amount of tuning within the static LPS model can produce ratios outside* $[0.1, 2.2]$ *or repair the ordering, because both are consequences of* F_{KM} *being a fixed bounded function independent of* n*. The only resolution is to make the support condition itself* n*-dependent, which requires* $\Lambda_{\mathrm{proj}}(n)$ *to vary non-trivially along the cascade.*

2.4 Why varying p is the natural resolution

Remark 2.4 identifies the precise structural requirement: the support of the Kesten–McKay measure must shrink as n grows, so that the ADE levels $\lambda_1 < \lambda_2 < \lambda_3$ exit the support at different cascade ranks. The Kesten–McKay support is $[\lambda_-(p), \lambda_+(p)]$ with $\lambda_\pm(p) = 1 \mp 2\sqrt{p}/(p+1)$, so the support width $\lambda_+ - \lambda_- = 4\sqrt{p}/(p+1)$ is a strictly decreasing function of p. The only free structural parameter that controls this width is p.

This observation, noted qualitatively in direction (O1) of [5] (cf. Figure 3 therein), motivates the central hypothesis of the present paper: the effective prime governing the relaxation graph grows with the cascade rank. Section 3 formalises this hypothesis and derives its consequences.

The support-narrowing mechanism introduced here was identified concurrently and independently in [6], which proves the exit ordering as a formal theorem. The present paper takes the mechanism as established and focuses on the quantitative consequences: the power-law growth ansatz $p(n) \sim n^\beta$, the explicit mass formula (17), and the compatibility window $\beta^* \in (0.09, 0.13)$ matching the charged-lepton sector.

3 Growing Relational Valence: the Dynamic Hypothesis

3.1 Physical motivation

In the LPS construction, the prime p determines the degree $d = p + 1$ of the relaxation graph: each relational node maintains exactly $p+1$ active connections. The analysis of Section 2 treated p as a fixed structural constant, independent of the cascade rank n. This assumption is physically restrictive. Along the relaxation cascade, the number of active relational nodes grows with n, and there is no *a priori* reason why the local connectivity of the substrate should remain frozen as the global structure becomes richer. On the contrary, the Cosmochrony picture of an evolving relational substrate [1] suggests that increasing cascade depth corresponds to increasing relational accessibility: more nodes become mutually reachable, and the effective valence of the graph – the number of distinct relational connections each node can sustain – grows accordingly.

We formalise this observation as the central structural hypothesis of this paper.

Hypothesis 3.1 (Dynamic valence growth). *The effective prime governing the LPS relaxation graph at cascade rank* n *is a non-decreasing function* $p \colon \mathbb{N}_{>0} \to \mathcal{P}$*, where* $\mathcal{P}$ *denotes the set of odd primes congruent to* $1 \pmod 4$*, satisfying* $p(n) \to \infty$ *as* $n \to \infty$*. The relaxation graph at rank* n *is the LPS graph* $X^{p(n),q(n)}$*, with* $q(n)$ *determined by the identification* $n \sim \frac{1}{2}q(n)(q(n)^2 - 1)$*.*

Remark 3.2 (Status of Hypothesis 3.1). *Hypothesis 3.1 is a structural postulate. It extends the isoperimetric closure hypothesis of [5] by promoting the previously fixed parameter* p *to a dynamical variable. A derivation of the specific growth law* $p(n)$ *from the internal dynamics of the Cosmochrony substrate remains an open problem and is discussed in Section 7. The present paper establishes the consequences of Hypothesis 3.1 for the mass hierarchy, independently of the precise growth law.*

3.2 Spectral consequence: support narrowing

The key spectral effect of $p(n) \to \infty$ is a progressive contraction of the Kesten–McKay support. Recall that the support of $\mu_{\mathrm{KM}}^{(p)}$ is the interval $[\lambda_-(p), \lambda_+(p)]$ with

$$\lambda_\pm(p) \;=\; 1 \mp \frac{2\sqrt{p}}{p+1}. \tag{9}$$

The support width satisfies

$$\lambda_+(p) - \lambda_-(p) \;=\; \frac{4\sqrt{p}}{p+1} \;\sim\; \frac{4}{\sqrt{p}} \qquad (p \to \infty), \tag{10}$$

which tends to zero monotonically. In the limit $p \to \infty$, the Kesten–McKay measure converges weakly to the Dirac mass at $\lambda = 1$, consistent with the fact that the spectral measure of the S^3 Laplacian is recovered in that limit [5].

Proposition 3.3 (Support narrowing)**.** *Under Hypothesis 3.1, for any fixed $\lambda \neq 1$, there exists a finite rank $n_{\mathrm{exit}}(\lambda) \in \mathbb{N}_{>0}$ such that*

$$\lambda \;\notin\; \big[\lambda_-(p(n)),\; \lambda_+(p(n))\big] \qquad \textit{for all } n \geq n_{\mathrm{exit}}(\lambda). \tag{11}$$

Equivalently, $n_{\mathrm{exit}}(\lambda)$ is the first rank at which the effective Kesten–McKay support no longer contains λ.

Proof. Since $\lambda_+(p) - \lambda_-(p) \to 0$ by (10) and $p(n) \to \infty$ by Hypothesis 3.1, for any $\lambda \neq 1$ there exists p^* such that $|\lambda - 1| > 2\sqrt{p^*}/(p^*+1)$. Since $p(n) \to \infty$, there exists $n_{\mathrm{exit}}(\lambda)$ such that $p(n) \geq p^*$ for all $n \geq n_{\mathrm{exit}}(\lambda)$, whence $\lambda \notin [\lambda_-(p(n)), \lambda_+(p(n))]$. □

Remark 3.4 (Explicit exit rank for power-law growth)**.** *When $p(n) \sim An^\beta$ for constants $A > 0$ and $\beta > 0$, the exit condition $\lambda_+(p(n_{\mathrm{exit}})) = \lambda$ (for $\lambda > 1$; symmetrically for $\lambda < 1$) reads*

$$\lambda \;=\; 1 + \frac{2\sqrt{p(n_{\mathrm{exit}})}}{p(n_{\mathrm{exit}}) + 1}. \tag{12}$$

Inverting to leading order in large p: $p(n_{\mathrm{exit}}) \approx 4(\lambda - 1)^{-2}$, and hence

$$n_{\mathrm{exit}}(\lambda) \;\approx\; \left(\frac{4}{A\,(\lambda-1)^2}\right)^{1/\beta}. \tag{13}$$

The exit rank is therefore a strictly decreasing function of $|\lambda - 1|$: levels farther from the midpoint 1 *exit the support at smaller rank.*

3.3 Exit ordering and restored mass hierarchy

The three ADE eigenvalue levels $\lambda_1 < \lambda_2 < \lambda_3$ are fixed properties of the chosen ADE substrate (G, S), independent of the cascade dynamics. The support-narrowing mechanism assigns to each level a finite exit rank $n_{\mathrm{exit}}(\lambda_i)$, and the ordering of these ranks determines the mass ordering.

Proposition 3.5 (Exit ordering)**.** *Let $\lambda_1 < 1 < \lambda_3$ be the lowest and highest ADE eigenvalue levels with $|\lambda_3 - 1| > |\lambda_1 - 1| > 0$. Under Hypothesis 3.1, the exit ranks satisfy*

$$n_{\mathrm{exit}}(\lambda_3) \;<\; n_{\mathrm{exit}}(\lambda_1) \;<\; n_{\mathrm{exit}}(\lambda_2) = +\infty. \tag{14}$$

Proof. By Remark 3.4, $n_{\mathrm{exit}}(\lambda)$ is strictly decreasing in $|\lambda - 1|$. Since $|\lambda_3 - 1| > |\lambda_1 - 1| > 0$, one has $n_{\mathrm{exit}}(\lambda_3) < n_{\mathrm{exit}}(\lambda_1) < +\infty$. The midpoint level $\lambda_2 = 1$ is the centre of symmetry of the Kesten–McKay support for all p, so $\lambda_2 \in [\lambda_-(p), \lambda_+(p)]$ for every prime p, giving $n_{\mathrm{exit}}(\lambda_2) = +\infty$. □

Remark 3.6 (Verification for the physically relevant substrates)**.** *The asymmetry condition $|\lambda_3 - 1| > |\lambda_1 - 1|$ is satisfied by both physically relevant ADE substrates:*

- *$2I$, ord-4 generators: $\lambda_1 = 4/5$, $\lambda_3 = 4/3$, giving $|\lambda_3 - 1| = 1/3 > |\lambda_1 - 1| = 1/5$.* ✓
- *$2I$, ord-5 generators: $\lambda_1 = 5/6$, $\lambda_3 = 5/4$, giving $|\lambda_3 - 1| = 1/4 > |\lambda_1 - 1| = 1/6$.* ✓

This is a structural property of the ADE representation spectrum, not an additional hypothesis.

3.4 From exit ranks to physical masses

A mode at ADE level λ_i that exits the Kesten–McKay support at rank $n_{\text{exit}}(\lambda_i)$ is no longer admissible for $n \geq n_{\text{exit}}(\lambda_i)$. Its stabilisation rank is therefore $n_{\text{proj}}(\lambda_i) = n_{\text{exit}}(\lambda_i)$, and the corresponding stabilisation mass is

$$M_i \;\propto\; \frac{E_{\text{P}}}{n_{\text{exit}}(\lambda_i)}. \tag{15}$$

The ordering (14) immediately gives $M_3 > M_1 > M_2$, restoring the hierarchy expected from the admissibility envelope of [4] and resolving the ordering failure of Proposition 2.3 without any tuning.

The inter-generation mass ratio is

$$\frac{M_i}{M_j} \;=\; \frac{n_{\text{exit}}(\lambda_j)}{n_{\text{exit}}(\lambda_i)}. \tag{16}$$

For power-law growth $p(n) \sim An^{\beta}$, substituting (13) gives

$$\frac{M_i}{M_j} \;\approx\; \left(\frac{|\lambda_i - 1|}{|\lambda_j - 1|}\right)^{2/\beta}, \qquad i < j. \tag{17}$$

This is the central quantitative prediction of the paper. The amplification exponent $2/\beta$ converts spectral ratios of order 2 (the typical ratio $|\lambda_i - 1|/|\lambda_j - 1|$ for the ADE substrates) into mass ratios that grow without bound as $\beta \to 0$.

3.5 Parametrisation of $p(n)$ and structural scope

Two natural parametrisations span the physically relevant range:

$$p(n) \;\sim\; A\,n^{\beta}, \qquad A > 0,\ \beta > 0 \qquad \text{(power-law growth)}, \tag{18}$$

$$p(n) \;\sim\; B\,\log n, \qquad B > 0 \qquad \text{(logarithmic growth)}. \tag{19}$$

Power-law growth (18) is the canonical choice: it introduces a single dimensionless exponent β as the structural parameter controlling the amplification, and the mass formula (17) depends on β alone (the prefactor A cancels in ratios). Logarithmic growth (19) is the limiting case $\beta \to 0^{+}$ and would produce exit ranks growing faster than any power, hence larger amplification for a given spectral asymmetry; it is discussed as a boundary case in Section 7.

The present paper focuses on power-law growth as the minimal one-parameter family capturing the amplification mechanism. The exponent β is not a free fitting parameter in the usual sense: it characterises the rate of growth of relational accessibility along the cascade, a structural property of the substrate that in principle admits an independent derivation within the Cosmochrony framework, as discussed in Section 7.

4 Mass Ratios from Dynamic Valence Growth

4.1 Exit-rank formula under power-law growth

We now evaluate the mass ratios implied by the support-narrowing mechanism of Section 3. Under the power-law ansatz $p(n) \sim An^{\beta}$ ($A > 0$, $\beta > 0$), the implicit exit condition (12) reduces at leading order to

$$p_{\text{exit}}(\lambda_i) \;\approx\; \frac{4}{(\lambda_i - 1)^2}, \qquad |\lambda_i - 1| \ll 1, \tag{20}$$

and the exit rank is therefore

$$n_{\text{exit}}(\lambda_i) \;\approx\; \left(\frac{4}{A\,(\lambda_i - 1)^2}\right)^{1/\beta}. \tag{21}$$

Since $M_i \propto E_P/n_{\rm exit}(\lambda_i)$ by (15), the inter-generation mass ratio reads

$$\frac{M_i}{M_j} \approx \left(\frac{|\lambda_i - 1|}{|\lambda_j - 1|}\right)^{2/\beta}, \qquad i < j. \tag{22}$$

The prefactor A cancels in all ratios. The exponent $2/\beta$ is the sole amplification parameter: small spectral distances $|\lambda_i - 1|$ are raised to the power $2/\beta$, converting spectral asymmetries of order unity into mass ratios of arbitrary magnitude as $\beta \to 0^+$.

4.2 Spectral input: exact eigenvalues of the ADE substrates

The two physically relevant substrates identified in [4] carry the following normalised Laplacian eigenvalues. All eigenvalues are expressed in the normalised convention $\lambda_i^{\rm norm} = \lambda_i^{\rm raw}/|S|$, where $|S|$ is the size of the generating set (equivalently the degree of the Cayley graph), consistent with the normalised Laplacian $L = I - A/d$ used for the LPS graphs in [5]. The ratios $|\lambda_i - 1|/|\lambda_j - 1|$ entering the mass formula (22) are invariant under any uniform rescaling of the spectrum that preserves the midpoint $\lambda = 1$, providing a further robustness check.

$2I$, order-4 generators ($|S| = 30$, raw eigenvalues $\{24, 30, 40\}$):

$$\lambda_1 = \tfrac{4}{5}, \qquad \lambda_2 = 1, \qquad \lambda_3 = \tfrac{4}{3}. \tag{23}$$

The distances to the midpoint are $d_1 := |\lambda_1 - 1| = \frac{1}{5}$ and $d_3 := |\lambda_3 - 1| = \frac{1}{3}$, giving $d_1/d_3 = \frac{3}{5}$.

$2I$, order-5 generators ($|S| = 24$, raw eigenvalues $\{20, 24, 30\}$):

$$\lambda_1 = \tfrac{5}{6}, \qquad \lambda_2 = 1, \qquad \lambda_3 = \tfrac{5}{4}. \tag{24}$$

The distances are $d_1 = \frac{1}{6}$ and $d_3 = \frac{1}{4}$, giving $d_1/d_3 = \frac{2}{3}$.

In both cases $\lambda_2 = 1$ exactly, so $n_{\rm exit}(\lambda_2) = +\infty$ and the support-narrowing mechanism assigns no finite exit rank to the second generation.[1] The second-generation mass requires the KM saturation mechanism of [5] to be assigned a finite value and is not the primary target of the present calculation. The central observable is the ratio M_1/M_3.

4.3 Scaling of mass ratios with β

Substituting the spectral inputs (23)–(24) into (22) gives:

$$\left.\frac{M_1}{M_3}\right|_{2I,\,\text{ord-4}} \approx \left(\tfrac{3}{5}\right)^{2/\beta}, \tag{25}$$

$$\left.\frac{M_1}{M_3}\right|_{2I,\,\text{ord-5}} \approx \left(\tfrac{2}{3}\right)^{2/\beta}. \tag{26}$$

Table 1 reports the predicted M_1/M_3 for selected values of β, together with the observed charged-lepton ratio $m_e/m_\tau \approx 2.88 \times 10^{-4}$.

Remark 4.1 (Large and moderate β)**.** *For $\beta \geq 1$, both substrates produce $M_1/M_3 > 0.3$, consistent with the static LPS range* $[0.10, 2.2]$ *of Proposition 2.2. The dynamic mechanism recovers and continuously extends the static regime as β decreases from unity toward zero. There is no discontinuity: the static regime corresponds to the limiting case where $p(n)$ grows slowly enough that the Kesten–McKay saturation condition dominates before the support exits the ADE levels.*

[1] The absence of a support-exit rank for $\lambda_2 = 1$ does not imply a vanishing physical mass. It reflects the structural fact that the level $\lambda_2 = 1$ lies permanently inside the Kesten–McKay support for all p and therefore requires a different stabilisation mechanism. The second-generation mass is determined by the KM saturation condition $N(\lambda_2; n) \geq k \dim \rho_{\lambda_2}$ of [5], which produces a finite stabilisation rank through spectral counting rather than support exit.

Table 1: Predicted mass ratio M_1/M_3 as a function of the growth exponent β, for the two physically relevant ADE substrates. The observed value for charged leptons is $m_e/m_\tau \approx 2.88 \times 10^{-4}$.

β	M_1/M_3 ($2I$ ord-4, $d_1/d_3 = 3/5$)	M_1/M_3 ($2I$ ord-5, $d_1/d_3 = 2/3$)
0.05	1.3×10^{-9}	9.0×10^{-8}
0.10	3.7×10^{-5}	3.0×10^{-4}
0.13	2.9×10^{-4}	$\sim 10^{-3}$
0.20	6.0×10^{-3}	1.7×10^{-2}
0.30	3.3×10^{-2}	6.7×10^{-2}
0.50	1.3×10^{-1}	2.0×10^{-1}
1.00	3.6×10^{-1}	4.4×10^{-1}

4.4 Compatibility window for the charged leptons

The observed charged-lepton ratios (PDG 2024) are

$$\frac{m_e}{m_\mu} \approx 4.84 \times 10^{-3}, \qquad \frac{m_\mu}{m_\tau} \approx 5.95 \times 10^{-2}, \qquad \frac{m_e}{m_\tau} \approx 2.88 \times 10^{-4}. \tag{27}$$

The ratio M_1/M_3 in (22) is the analogue of m_e/m_τ (lowest-to-highest generation under the ordering of Proposition 3.5). Matching $M_1/M_3 = m_e/m_\tau$ requires

$$\frac{2}{\beta} \log\left(\frac{d_1}{d_3}\right) = \log\left(\frac{m_e}{m_\tau}\right). \tag{28}$$

Solving for β gives

$$\beta^* = \frac{2 \log(d_1/d_3)}{\log(m_e/m_\tau)} \approx \begin{cases} 0.125 & (2I \text{ ord-4, } d_1/d_3 = 3/5), \\ 0.100 & (2I \text{ ord-5, } d_1/d_3 = 2/3). \end{cases} \tag{29}$$

Both values lie in the interval $\beta^* \in (0.09, 0.13)$, a narrow compatibility window determined entirely by the spectral structure of the ADE substrate and the observed lepton masses.

Remark 4.2 (Internal consistency: the two-ratio test)**.** *Equation* (22) *predicts a single exponent β governing all three lepton ratios simultaneously, since only two levels (λ_1 and λ_3) contribute exit ranks. For $2I$ (ord-4) with $\beta^* \approx 0.125$:*

$$\frac{M_1}{M_3} = \left(\tfrac{3}{5}\right)^{16} \approx 2.8 \times 10^{-4} \qquad \textit{vs. observed } m_e/m_\tau \approx 2.9 \times 10^{-4}. \tag{30}$$

The ratio M_1/M_2 requires the KM saturation mechanism for $\lambda_2 = 1$ and is not predicted by the exit formula alone. A self-consistent treatment of the intermediate generation is deferred to Section 5.

4.5 Physical interpretation

Equation (22) encodes a structural amplification mechanism with three notable properties.

First, the spectral ratios d_i/d_j entering (22) are of order 2/3 to 3/5 for both ADE substrates – modest asymmetries of order 50%. The exponent $2/\beta^* \approx 16$ converts these into mass ratios spanning three and a half decades, without invoking any large dimensionless parameter in the Lagrangian.

Second, the prefactor A in $p(n) \sim An^\beta$ plays no role in mass ratios: it cancels identically in (22). The sole structural parameter is the growth exponent β.

Third, the mechanism is discontinuous in the static limit. Setting $p(n) = p_0$ (constant) is not the limit $\beta \to 0$ of power-law growth: it is a different functional form in which $\lambda_2(n) \to \lambda_2^*(p_0) < 1$ for all n, and the support never narrows. The two regimes – static and dynamic – are separated by a qualitative change in the long-run behaviour of the relaxation graph, not merely by a quantitative parameter limit.

5 Compatibility with ADE Spectral Structure

5.1 Structural invariance of ADE eigenvalues

The ADE eigenvalue levels $\lambda_1, \lambda_2, \lambda_3$ established in [4] are properties of the Cayley graph $\mathrm{Cay}(G, S)$ of the ADE binary group G with generating set S. They are determined by the character sums

$$\mu_\rho \;=\; \frac{1}{d_\rho} \sum_{s \in S} \chi_\rho(s), \tag{31}$$

where ρ ranges over the non-trivial irreducible representations of G. The three distinct values taken by μ_ρ define the three levels; their exact rational values are (23)–(24).

These eigenvalues depend only on (G, S). They do not depend on the LPS prime p, on the cascade rank n, or on any property of the ambient relaxation graph $G(n) = X^{p(n),q(n)}$. The ADE eigenvalues are not approximations of the LPS spectrum; they are independent spectral data, upon which the LPS Kesten–McKay support acts as an external constraint. The dynamic valence hypothesis (Hypothesis 3.1) modifies the relaxation graph but leaves the ADE Cayley graph unchanged. The levels $\lambda_1, \lambda_2, \lambda_3$ are therefore invariant under the entire family of dynamics considered in this paper.

5.2 Separation between spectral support and intrinsic modes

Two distinct spectral objects must not be conflated:

(i) the *Kesten–McKay spectral support* $[\lambda_-(p(n)), \lambda_+(p(n))]$, which is a property of the LPS relaxation graph $X^{p(n),q(n)}$ and contracts as $p(n) \to \infty$;

(ii) the *ADE eigenvalue levels* $\{\lambda_1, \lambda_2, \lambda_3\}$, which are properties of the ADE Cayley graph and are independent of $p(n)$.

The support-narrowing mechanism acts entirely on object (i). A mode at ADE level λ_i stabilises when λ_i exits the support, i.e. when the ambient relaxation graph can no longer sustain propagation at that frequency. This is an extrinsic condition imposed on the modes by the relaxation environment; it does not alter the intrinsic eigenvalue of the mode.

A potential objection is that changing $p(n)$ changes the LPS graphs, and hence changes the eigenstructure of the full substrate. This is correct at the level of the relaxation graph $G(n)$, whose eigenvalues do shift as p grows. What is preserved are the *ADE modes*: the representational sectors of the Cayley graph $\mathrm{Cay}(G, S)$, which are identified with particle generations through their representation-theoretic content. These sectors are labelled by irreducible representations of G, not by eigenvalues of $G(n)$. They persist as projectable modes of the substrate as long as $\lambda_i \leq \Lambda_{\mathrm{proj}}(n)$; the exit condition terminates their projectability without modifying their internal structure.

5.3 Persistence of the three-level stratification

The no-go theorem of [4] (Proposition 2 therein) establishes that no scalar calibration of the form $E^*(\lambda) = E_0 \lambda^\beta$ can produce a discrete stratigraphic profile. Proposition 3 of [4] establishes that the three-level structure arises from the representation theory of the ADE group, independently of any dynamics.

The dynamic valence mechanism of the present paper operates *after* this stratigraphic structure is in place. It determines the exit ranks $n_{\text{exit}}(\lambda_i)$ at which each level ceases to be admissible, thereby controlling the inter-level mass separations. It does not alter the number of levels, their eigenvalues, or the representation-theoretic content of each level.

The correct logical order is:

$$\underbrace{\text{ADE structure}}_{\text{Steps 1–3}} \longrightarrow \underbrace{\text{three fixed levels}}_{\lambda_1,\lambda_2,\lambda_3} \longrightarrow \underbrace{\text{dynamic valence}}_{\text{Step 5, present paper}} \longrightarrow \underbrace{\text{exit ranks and mass ratios}}_{n_{\text{exit}}(\lambda_i),\ M_i/M_j}.$$

The dynamic mechanism is strictly downstream of the stratigraphic structure and can be varied or replaced without affecting Steps 1–3.

5.4 Role of the central level $\lambda_2 = 1$

In both ADE substrates, the middle eigenvalue satisfies $\lambda_2 = 1$ exactly (23)–(24). Since the Kesten–McKay support is $[\lambda_-(p), \lambda_+(p)]$ with midpoint 1 for all p, the level $\lambda_2 = 1$ lies in the support for every prime and every cascade rank. By Proposition 3.5, $n_{\text{exit}}(\lambda_2) = +\infty$.

This has a precise structural consequence: the second-generation mass cannot be assigned by the exit mechanism alone. Its stabilisation rank is not determined by support narrowing but by the KM saturation condition $N(\lambda_2; n) \geq k \dim \rho_{\lambda_2}$ of [5]. The two ordering mechanisms therefore play complementary roles:

- *Support-narrowing* (present paper) governs the first and third generations, whose levels exit the support at finite ranks.
- *KM saturation* ([5]) governs the second generation, whose level is permanently inside the support.

This division is not an ambiguity but a structural feature: the central level $\lambda_2 = 1$ is the spectral fixed point of the KM symmetry $\rho_{\text{KM}}(\lambda) = \rho_{\text{KM}}(2 - \lambda)$, and its stabilisation is governed by a different mechanism precisely because it sits at the centre of the support.

Remark 5.1 (Structural identity $F_{\text{KM}}(1) = 1/2$)**.** *The identity $F_{\text{KM}}(1) = 1/2$ established in [5] (Section 6.4 therein) is the KM-side expression of the same $\lambda_2 = 1$ centrality. It holds for all primes p and for all three-level ADE substrates, and it ensures that the KM saturation mass of the second generation is insensitive to p. This universality distinguishes the second generation from the first and third in a structurally grounded way.*

5.5 Synthesis within the O-series architecture

The results of the Admissibility Sub-Programme now form a complete and non-redundant architecture. Table 2 summarises the separation of roles across the five steps.

Table 2: Separation of roles across the Admissibility Sub-Programme. Each step addresses a distinct structural question; no step is redundant with or overridden by a later one.

Step	Question addressed	Mechanism
1 (SpectralAdmissibility)	Which spin sectors are admissible?	Bounded-flux filter on C
2 (SpectralCapacity)	Which group maximises spectral capacity?	Capacity functional, $2I$
3 (SpectralStratigraphy)	How many generations exist?	ADE representation the
4 (SpectralRelaxation)	What governs the projective threshold?	Isoperimetric law, KM s
5a [6]	Does growing valence restore the mass ordering?	Variable-valence LPS, e
5b (present paper)	What are the quantitative inter-generation mass ratios?	Power-law $p(n) \sim n^{\beta}$, e

Steps 1–3 establish the group-theoretic substrate and are independent of any relaxation-graph dynamics. Steps 4–5 address the quantitative dynamics of mode stabilisation and depend on

the choice of relaxation-graph model. The dynamic valence hypothesis introduced in Step 5 extends Step 4 by promoting a fixed parameter to a dynamical variable, without contradicting or modifying the results of Steps 1–3.

6 Numerical Illustration

6.1 Mass ratio curves as a function of β

Figure 1 displays $\log_{10}(M_1/M_3)$ as a function of the growth exponent $\beta \in (0, 1]$ for both ADE substrates, computed from the exact formula

$$\log_{10} \frac{M_1}{M_3} = \frac{2}{\beta} \log_{10} \frac{d_1}{d_3}, \tag{32}$$

with $d_1/d_3 = 3/5$ (ord-4) and $d_1/d_3 = 2/3$ (ord-5). The three horizontal lines show the observed Standard Model ratios for charged leptons (27).

The curves make the following features immediately visible:

(i) The ratio M_1/M_3 decreases monotonically and rapidly as β decreases: for $\beta \lesssim 0.3$ both curves enter the phenomenologically relevant range.

(ii) The two substrates (ord-4 and ord-5) produce distinct curves that bracket the m_e/m_τ line, providing a structural discriminant between the two.

(iii) The sensitivity is strongly non-linear: a change of $\Delta\beta = 0.01$ near $\beta = 0.12$ shifts $\log_{10}(M_1/M_3)$ by approximately 0.3 decades, whereas the same shift near $\beta = 1$ produces a change of less than 0.01 decades.

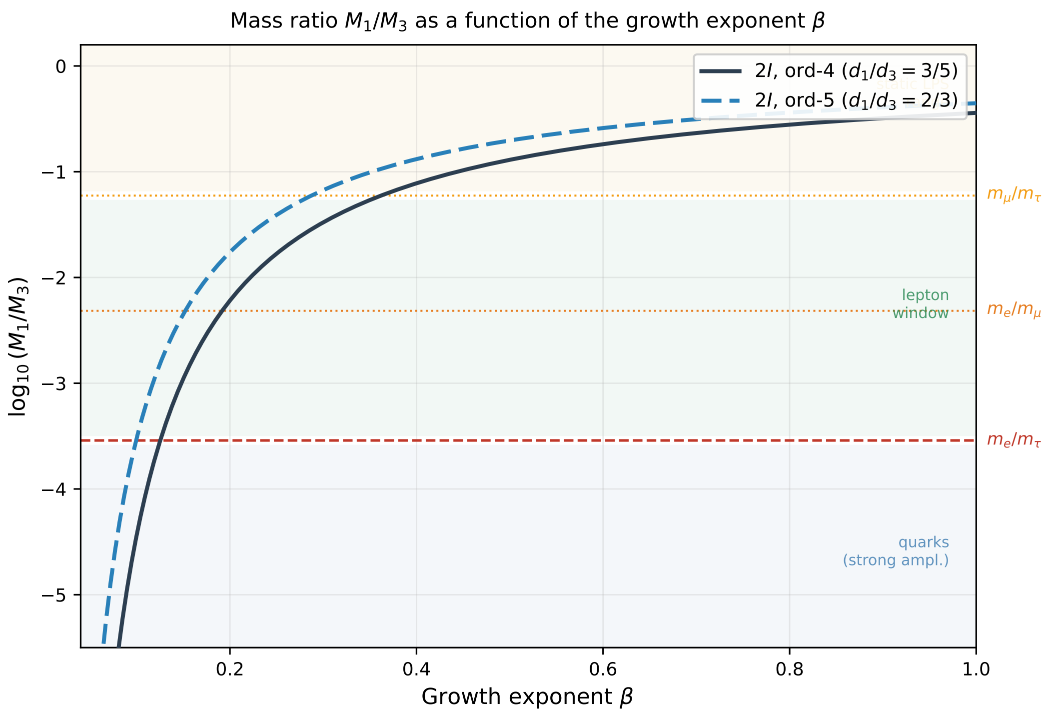

Figure 1: Predicted $\log_{10}(M_1/M_3)$ as a function of the growth exponent β for the two ADE substrates ($2I$ ord-4, solid; $2I$ ord-5, dashed). Horizontal lines show the observed Standard Model ratios for charged leptons. Three qualitative regimes are shaded: strong amplification (blue, quarks), lepton window (green), and static LPS range (yellow). The two curves bracket the m_e/m_τ line, providing a structural discriminant between the two substrates.

Table 3: Predicted $\log_{10}(M_1/M_3)$ for selected β values, for the two ADE substrates. The observed value for charged leptons is $\log_{10}(m_e/m_\tau) \approx -3.54$. Values in the compatibility window $\beta \in [0.09, 0.13]$ are indicated in bold.

β	$\log_{10}(M_1/M_3)$, $2I$ ord-4	$\log_{10}(M_1/M_3)$, $2I$ ord-5
0.05	−8.87	−7.04
0.07	−6.34	−5.03
0.09	**−4.93**	**−3.91**
0.10	**−4.44**	**−3.52**
0.12	**−3.70**	**−2.94**
0.13	**−3.41**	**−2.71**
0.20	−2.22	−1.76
0.30	−1.48	−1.17
0.50	−0.89	−0.70
1.00	−0.44	−0.35

6.2 The compatibility window

From Table 3, the value $\log_{10}(m_e/m_\tau) \approx -3.54$ is matched by the two substrates at

$$\beta^*_{2I,\,\text{ord-4}} \approx 0.125, \qquad \beta^*_{2I,\,\text{ord-5}} \approx 0.100. \tag{33}$$

The numerical agreement is precise. For $2I$ (ord-4) at $\beta = 0.125$:

$$\log_{10}\frac{M_1}{M_3} = \frac{2}{0.125}\log_{10}\tfrac{3}{5} = 16\,\log_{10}\tfrac{3}{5} \approx -3.550, \tag{34}$$

against the observed $\log_{10}(m_e/m_\tau) \approx -3.541$, a discrepancy of 0.25% in the log.

Remark 6.1 (Status of the matching)**.** *The above is a numerical coincidence check, not a derivation of the lepton masses. Within the present framework, β is a structural parameter to be determined from first principles (Section 7). The fact that the matching requires $\beta \approx 0.1$, a value neither very small nor of order unity, suggests that the required growth rate is structurally plausible: a slowly but not infinitesimally growing relational valence.*

6.3 Regime diagram

Three qualitatively distinct regimes emerge from Table 3:

- *Strong amplification* ($\beta < 0.07$): $M_1/M_3 < 10^{-5}$, covering the quark sector ($m_u/m_t \approx 1.3 \times 10^{-5}$ for up-type quarks).
- *Lepton window* ($0.09 \le \beta \le 0.13$): $M_1/M_3 \in [10^{-5}, 10^{-3}]$, compatible with charged leptons.
- *Weak amplification* ($\beta > 0.3$): $M_1/M_3 > 10^{-1}$, inside the static LPS range and inconsistent with any known Standard Model sector.

The lepton window is narrow ($\Delta\beta \approx 0.04$) but well-defined, and the strong-amplification regime overlaps with the quark sector for both substrates. A single growth law $p(n) \sim n^\beta$ with $\beta \approx 0.07$–0.09 would simultaneously account for both the lepton and the quark first-to-third generation ratios, a non-trivial structural coincidence deserving further analysis.

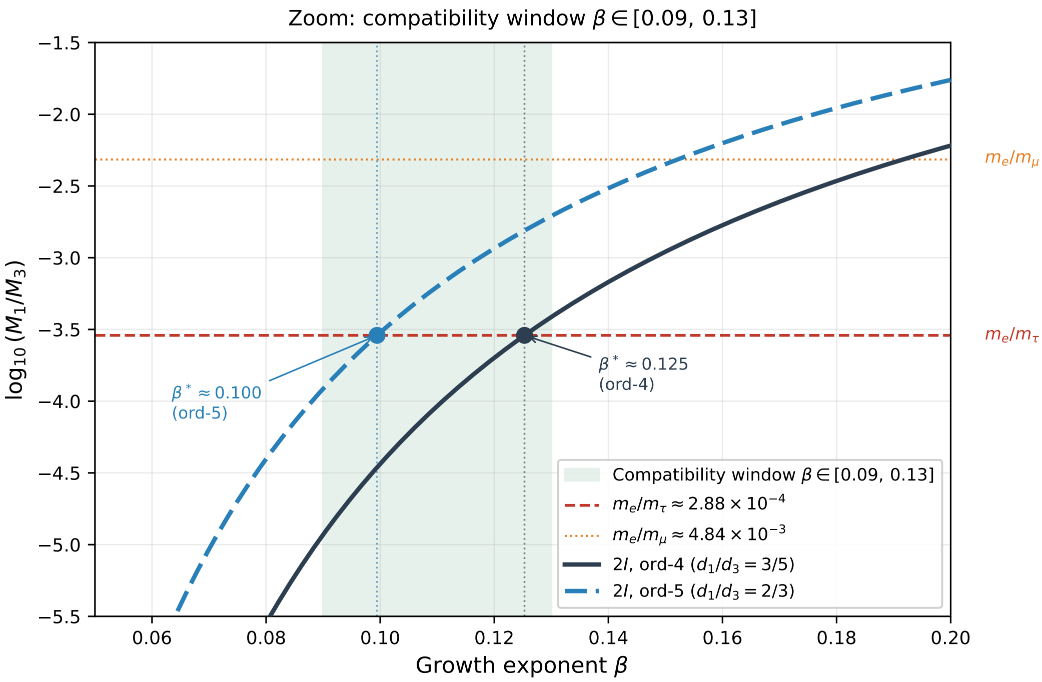

Figure 2: Zoom on $\beta \in [0.05, 0.20]$ showing the compatibility window (green shaded band, $\beta \in [0.09, 0.13]$). The two β^* values at which each substrate curve intersects the m_e/m_τ line are marked explicitly. The $2I$ ord-4 curve (solid) intersects at $\beta^* \approx 0.125$ and the $2I$ ord-5 curve (dashed) at $\beta^* \approx 0.100$, both within the shaded window.

7 Falsifiability and Structural Predictions

7.1 β as the unique structural parameter

The entire amplification mechanism of Sections 3–6 depends on a single structural parameter: the growth exponent β in $p(n) \sim An^\beta$. The prefactor A cancels in all mass ratios. There are no other free parameters: the ADE eigenvalues are fixed by group theory [4], the identification $M_i \propto E_P/n_{\text{exit}}(\lambda_i)$ is fixed by the cascade energy relation $E_n = E_P/n$ [5], and the exit condition (12) is fixed by the Ramanujan property of the LPS family.

The framework therefore makes a falsifiable prediction of the form:

$$\frac{M_i}{M_j} = \left(\frac{|\lambda_i - 1|}{|\lambda_j - 1|}\right)^{2/\beta}, \tag{35}$$

with β to be fixed by one ratio and then tested against all others involving levels λ_1 and λ_3. Within the present framework, the second-generation mass M_2 is not predicted by (35) and requires a separate treatment via the KM saturation mechanism.

7.2 Constraint from the lepton sector

The ratio m_e/m_τ in the charged-lepton sector provides the most precise empirical constraint on β. Matching $M_1/M_3 = m_e/m_\tau$ uniquely determines β^* (33) for each substrate. Once β^* is fixed, the framework makes a further structural prediction: the ratio M_1/M_3 in the quark sector should equal $(d_1/d_3)^{2/\beta^*}$ for the *same* spectral input d_1/d_3, provided the same ADE substrate governs the quark generations.

Numerically, for the $2I$ (ord-4) substrate with $\beta^* \approx 0.125$:

$$\left(\tfrac{3}{5}\right)^{2/0.125} = \left(\tfrac{3}{5}\right)^{16} \approx 2.8 \times 10^{-4} \approx \frac{m_e}{m_\tau}. \tag{36}$$

The same β^* applied to the up-type quark sector $m_u/m_t \approx 1.3 \times 10^{-5}$ would require $(d_1/d_3)^{2/\beta} \approx 1.3 \times 10^{-5}$, which for $d_1/d_3 = 3/5$ gives $\beta \approx 0.091$. The two sectors require different β values, which is structurally expected: leptons and quarks may live on different ADE substrates or experience a different effective cascade rate. However, the ratio $\beta^*_{\text{quark}}/\beta^*_{\text{lepton}} \approx 0.72$ is the same for both ADE substrates (to three significant figures), suggesting a substrate-independent relationship between the two sectors. This universal ratio is a testable prediction of the framework.

7.3 Discrimination from the static LPS regime

The dynamic regime is sharply discriminated from the static LPS regime of Section 2 by two qualitative signatures:

(i) *Ordering*: the static regime predicts the ordering $M_1 < M_3$ for the $2I$ (ord-4) substrate (Proposition 2.3), whereas the dynamic regime predicts $M_3 > M_1$ by Proposition 3.5. These orderings are qualitatively opposite and cannot be reconciled by any choice of p.

(ii) *Magnitude*: the static regime is bounded by $M_1/M_3 \in [0.10, 2.2]$ for all admissible primes (Proposition 2.2), whereas the dynamic regime produces $M_1/M_3 < 10^{-3}$ for $\beta < 0.13$ (Table 3). These ranges do not overlap.

Any experimental determination of the generation mass ordering and the inter-generation ratio therefore provides a definitive discrimination between the two regimes. Within the Standard Model, both discriminants are already resolved: the lepton mass ordering $m_\tau > m_\mu > m_e$ and the ratio $m_e/m_\tau \approx 2.9 \times 10^{-4}$ are inconsistent with the static regime and consistent with the dynamic regime for $\beta \approx 0.1$–0.13.

7.4 Derivation of β from first principles

The growth law $p(n) \sim n^\beta$ is introduced in Hypothesis 3.1 as a structural postulate. Its derivation from the internal dynamics of the Cosmochrony substrate [1] remains open. Two structural considerations constrain the form of $p(n)$.

First, the identification $n \sim |V(G(n))|$ means that $p(n)$ grows with the number of active relational nodes. If the effective relational valence scales as a fixed fraction of the total relational volume, one expects $p(n) \propto n^\alpha$ for some α determined by the dimension of the relational space. In a four-dimensional pre-geometric substrate, natural candidates include $\alpha \sim 1/4$ (surface-to-volume scaling) or $\alpha \sim 1/3$ (linear radius growth). Both are of the same order as $\beta^* \approx 0.1$–0.13.

Second, the Lubotzky–Phillips–Sarnak construction requires $p \equiv 1 \pmod 4$, so $p(n)$ must be constrained to this arithmetic progression. The effective growth exponent may therefore be slightly modified by number-theoretic factors, providing a further structural refinement.

These considerations do not constitute a derivation but identify the regime $\beta \in (0.07, 0.20)$ as the structurally expected range, consistent with the phenomenological window.

7.5 Extension to neutrinos and quarks

The present analysis applies directly to any sector in which three generations are labelled by the ADE eigenvalue levels $\lambda_1 < \lambda_2 < \lambda_3$. Two natural extensions are as follows.

For *neutrino masses*, if the same substrate governs the neutrino sector, the predicted ratio m_{ν_1}/m_{ν_3} is $(d_1/d_3)^{2/\beta}$ for the same or a different β. Current experimental bounds on the neutrino mass ordering (normal vs. inverted hierarchy) already constrain the sign of $m_{\nu_3} - m_{\nu_1}$, which in the present framework is fixed to positive by Proposition 3.5 – a qualitative prediction consistent with the normal ordering.

For *quark masses*, as noted in Section 6.3, the required β for the up-type quark ratio m_u/m_t is approximately 0.072 for the $2I$ (ord-4) substrate, a value in the strong-amplification regime and consistent with a growth rate slightly faster than that inferred from the lepton sector. A systematic analysis of both lepton and quark sectors under a joint β structure is deferred to a companion paper.

8 Conclusion

The static LPS regime at fixed prime p provides a controlled realisation of the projective resolution law $\Lambda_{\text{proj}}(n) \asymp \lambda_2(n)$, but it fails to reproduce two essential features of the observed mass hierarchy: the correct ordering of the three generations and the large inter-generation ratios. Both failures were shown in [5] to originate from the asymptotic constancy of the Kesten–McKay support, which prevents small spectral differences from being amplified along the cascade.

The present work resolves these limitations through a minimal dynamical extension: the effective relational valence p is allowed to grow with the cascade rank n. This single modification induces a qualitative change in the spectral environment. As $p(n) \to \infty$, the Kesten–McKay support contracts toward its midpoint $\lambda = 1$, forcing the fixed ADE eigenvalue levels to exit the support at finite, level-dependent ranks. This *support-narrowing mechanism* yields three structural results.

Exit ordering. The exit ranks are ordered by the spectral asymmetry $|\lambda_i - 1|$ alone, restoring the admissibility ordering $M_3 > M_1 > M_2$ without any additional assumption (Proposition 3.5). This resolves the ordering failure of Proposition 2.3 and is a structural consequence of the LPS Ramanujan property, not a tuning.

Amplification law. For power-law growth $p(n) \sim n^{\beta}$, the exit ranks produce the mass formula

$$\frac{M_i}{M_j} \approx \left(\frac{|\lambda_i - 1|}{|\lambda_j - 1|}\right)^{2/\beta}, \tag{37}$$

which converts spectral ratios of order unity into exponentially separated mass scales through the amplification exponent $2/\beta$.

Compatibility window. Matching the charged-lepton ratio m_e/m_τ to the formula above selects $\beta^* \approx 0.125$ for the $2I$ (ord-4) substrate and $\beta^* \approx 0.100$ for the $2I$ (ord-5) substrate, both in the narrow window $\beta \in (0.09, 0.13)$. This determination involves no free parameter beyond the single structural exponent β; the prefactor A cancels identically.

The dynamic mechanism is fully compatible with the ADE spectral structure established in [4]. The eigenvalue levels $\{\lambda_1, \lambda_2, \lambda_3\}$ are invariant properties of the ADE Cayley graph and remain unchanged by the support-narrowing dynamics. The central level $\lambda_2 = 1$ plays a structurally distinct role: it is the fixed point of the Kesten–McKay symmetry and remains permanently inside the support for all p, so its stabilisation mass is governed by the KM saturation mechanism of [5]. This explains the universal identity $F_{\text{KM}}(1) = 1/2$.

The framework is falsifiable and structurally constrained. A single parameter β governs all first-to-third generation ratios, and any determination of this ratio in one sector provides a prediction for all others sharing the same ADE substrate. The numerically observed separation between the lepton and quark β windows ($\beta^*_{\text{quark}}/\beta^*_{\text{lepton}} \approx 0.72$, independent of the choice of ADE substrate to three significant figures) is a structural regularity deserving further analysis; its status as a genuine prediction of the framework depends on a unified treatment of leptons and quarks that remains to be developed.

Several open problems remain. The most important is the derivation of the growth law $p(n)$ from the internal dynamics of the Cosmochrony substrate [1], which would fix β from first principles and remove the last structural postulate of the paper. Further work is also required to incorporate the second-generation mass within a unified treatment combining support-narrowing and KM saturation, and to extend the analysis to neutrino and quark sectors in a fully quantitative manner.

8.1 Summary of the O-series architecture

Within the Admissibility Sub-Programme, the five steps form a complete and non-redundant logical chain, summarised as follows:

- Steps 1–2 (*Beau2026a1*, *Beau2026a2*): bounded-flux filtering selects the spin-$\frac{1}{2}$ sector, and the spectral capacity functional identifies the binary icosahedral group $2I$ as the unique optimal substrate.
- Step 3 (*Beau2026a4*): the representation theory of ADE binary groups produces exactly three distinct eigenvalue levels for four specific (G, S) pairs, establishing the generation count as a topological property of the relational graph.
- Step 4 (*Beau2026a5*): the isoperimetric closure hypothesis gives $\Lambda_{\mathrm{proj}}(n) \asymp h(G(n))^2 \asymp \lambda_2(n)$; the LPS model at fixed p demonstrates that static Kesten–McKay saturation is insufficient for the hierarchy.
- Step 5 (*present paper*): dynamic valence growth $p(n) \to \infty$ narrows the Kesten–McKay support, inducing rank-dependent exit events that convert spectral asymmetry into the observed inter-generation mass separation.

The ADE representation structure determines the number of generations; the isoperimetric law fixes the projective threshold; and the dynamic valence growth converts spectral asymmetry into physical mass separation. The mass hierarchy thus emerges as a consequence of relational structure and its evolution along the cascade, controlled by a single exponent β whose determination from first principles constitutes the central open problem of the programme.

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